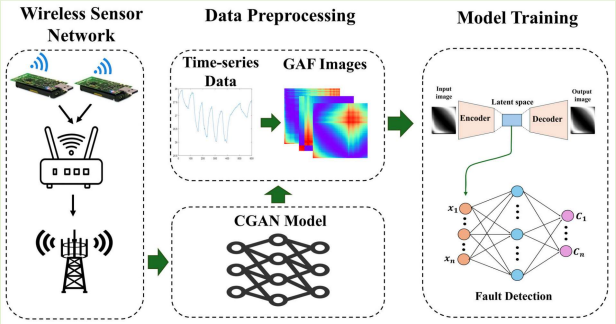
Robust Sensor Fault Detection in Wireless Sensor Networks Using a Hybrid Conditional Generative Adversarial Networks and Convolutional Autoencoder

Rehan Khan, Umer Saeed, and Insoo Koo

***Abstract*—In the rapidly growing realm of the Internet of Things (IoT), reliance on sensor-generated data has become crucial for the operation of multiple services and systems. As essential components of these systems, wireless sensor networks (WSNs) are installed in a wide range of diverse and often harsh environments. However, these networks are highly prone to a range of faults, including software bugs, communication failures, and hardware malfunctions. Such issues can lead data to data being transmitted incorrectly, endangering the security, reliability, and economic stability of the systems they support. Addressing the challenge of sensor fault detection, we propose a novel hybrid technique to enhance the classification of sensor fault data in WSNs. Our method leverages a publicly available dataset of temperature sensor readings to generate synthetic data by using conditional generative adversarial networks (GANs). These synthetic samples closely resemble common temperature sensor data despite the introduction of artificial sensor faults in WSNs, including hardover, drift, spike, erratic, and stuck faults. In order to capture the temporal dependencies in time-series data, we transform the sensor readings into Gramian angular field (GAF) images, retaining the temporal structure. These GAF images are then processed using a convolutional autoencoder (CAE) to extract rich feature representations, followed by a three-layer artificial neural network (ANN) for the multiclass classification of sensor faults. Our proposed method not only addresses the challenges of data scarcity and imbalance but also enhances accuracy in sensor fault detection. The proposed method demonstrates high accuracy, $F1$-score, recall, and sensitivity, achieving 95.93%, 95.84%, 95.88%, and 95.88%, respectively.**



***Index Terms*—Conditional generative adversarial networks (Conditional GANs), convolutional autoencoder (CAE), Gramian angular field (GAF), sensor faults, wireless sensor networks (WSNs).**

I. Introduction

IN RECENT years, the integration of the Internet of Things (IoT) and cyber-physical systems (CPSs) has revolutionized various sectors by bridging the physical and digital worlds through smart devices and sensors. These technologies have enabled the development of intelligent systems, offering new opportunities for automation, monitoring, and interactivity across diverse industries. IoT and CPS technologies have made common things into smart entities that can gather and transmit data by embedding sensing and communication capabilities within them [1]. Wireless sensor networks (WSNs), advanced communication protocols, and smart systems enabling real-time data gathering and transmission to central nodes are the driving forces behind this revolution [2].

The integration of IoT and WSNs has proven to be transformative across sectors such as agriculture, energy, and healthcare, demonstrating substantial improvements in efficiency, monitoring, and decision-making. In agriculture, IoT-based frameworks enable large-scale monitoring of environmental parameters such as temperature, soil moisture, and nutrient levels [3], facilitating precision farming techniques that optimize resource usage and boost yields, even under adverse conditions [4]. Similarly, the energy sector has

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witnessed advancements through smart grids that leverage IoT devices for real-time monitoring, enabling accurate demand forecasting, load management, and the detection of irregularities, including power theft [5], [6]. These innovations not only enhance service quality but also contribute to sustainability by supporting renewable energy adoption.

In healthcare, IoT technologies have enabled personalized monitoring systems [7], [8], providing real-time health data to medical professionals, improving chronic disease management, and facilitating timely interventions during emergencies [9]. Collectively, these advancements underscore the wide-ranging impact of IoT and WSNs in modernizing industries and addressing contemporary challenges.

Current research and implementation trends suggest that IoT-based services will keep expanding across a variety of industries. However, the integrity and reliability of the data transmitted by sensors remain crucial factors influencing the quality of these IoT applications [10]. Ensuring the accuracy and dependability of sensor data is a significant challenge for the efficacy of IoT systems. In real-world scenarios, sensors are often deployed in harsh and extreme environments, such as industrial facilities or remote natural settings [11]. These conditions expose sensors to potential faults caused by hardware aging, improper installation, or adverse environmental factors, including excessive temperatures or humidity [12].

In most IoT-based services, decision-making and control actions rely on sensor-gathered data. Consequently, malfunctioning sensors or faulty data, if not promptly identified and addressed, can compromise the accuracy and reliability of these systems, potentially leading to severe consequences [13]. To mitigate such risks, robust fault detection mechanisms are essential. Sensor faults are typically categorized into two broad categories: incipient faults, which develop gradually and are difficult to detect in their preliminary stages, and abrupt faults, which occur suddenly and are easier to identify [14].

Research into the identification and diagnosis of various types of faults in industrial facilities began in the 1980s [15]. Traditional sensor fault detection techniques are generally divided into three categories: model-based, knowledge-based, and data-driven methods [16]. Model-based approaches rely on analytical models derived from physical laws or system specifications, but developing precise models for complex systems is often labor-intensive, error-prone, and challenging to generalize, limiting their practical applicability [17]. Knowledge-based approaches leverage historical data and domain expertise, often incorporating fuzzy-rule-based or expert systems, as well as statistical techniques such as principal component analysis (PCA) and partial least squares (PLS), which have been widely employed for fault detection [18].

In contrast, data-driven approaches, enabled by advancements in machine learning (ML) and deep learning (DL), have emerged as highly effective tools for automatic fault detection. These methods operate directly on raw or preprocessed data to achieve high accuracy in detecting and classifying sensor faults. Moreover, the widespread adoption of networked sensors with cloud connectivity has further expanded access to large-scale sensor data, enhancing the scalability and applicability of data-driven techniques in sensor fault analysis.

In recent years, generative adversarial networks (GANs) have emerged as a groundbreaking approach for anomaly detection across various domains, including sensor fault detection. These methods are particularly effective in generating synthetic data that closely resemble real-world scenarios, thereby addressing critical challenges such as data scarcity and class imbalance [19]. GANs have demonstrated their adaptability in diverse fields, including medical diagnostics, communication systems, and industrial monitoring, where they efficiently handle tasks such as data augmentation and feature extraction. For example, in [20], GANs were employed to generate synthetic blood test data, improving anomaly detection for COVID-19 infection diagnosis and enhancing detection capabilities in cases with limited data availability. Similarly, Bouchenak et al. [21] explored a GAN-based semisupervised approach for modulation classification in multiple-input–multiple-output (MIMO) systems. These studies highlight the versatility of GANs in handling high-dimensional and complex datasets, showcasing their potential to effectively address various data challenges.

Ensuring the reliability and accuracy of WSNs is critical for their effective operation in real-world applications, as sensor faults can compromise decision-making and pose significant safety risks. Despite advancements in fault detection techniques, existing approaches often face challenges such as data scarcity and the complexity of modeling heterogeneous fault types. To address these limitations, this study introduces a hybrid approach for sensor fault detection, integrating ML and DL techniques with GAN-based data augmentation. By leveraging Conditional GANs, synthetic sensor readings closely mirror real-world scenarios, effectively mitigating data scarcity.

In addition, this study transforms time-series sensor readings into Gramian angular field (GAF) images to preserve temporal dependencies, which are crucial for effective feature extraction. The proposed framework utilizes a convolutional autoencoder (CAE) for feature extraction, followed by a three-layer artificial neural network (ANN) for classification. This approach provides a robust solution for identifying and diagnosing multiple fault types, including hardover, drift, spike, erratic, and stuck conditions. It ensures enhanced fault detection accuracy and adaptability across diverse operational environments.

The main contributions of this study are as follows.

1) This research introduces a significant advancement by utilizing Conditional GANs to generate synthetic data, effectively addressing the challenges associated with data scarcity. By generating synthetic data that closely mimics real-world sensor readings, the proposed approach enhances the reliability and robustness of fault detection systems in WSNs.
2) We artificially inject a range of common sensor fault readings (including hardover, drift, spike, erratic, and stuck faults) into normal temperature sensor readings from a multihop WSN scenario. This process enables the generation of realistic faulty sensor data, facilitating

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comprehensive testing and validation of fault detection models.

3) Time-series sensor readings are transformed into GAF images, preserving inherent temporal relationships and highlighting critical patterns. These images are further processed using a CAE, which compresses and reconstructs the data, effectively filtering noise and retaining critical spatial and temporal features. This integrated approach significantly enhances feature extraction, enabling precise fault classification.
4) The proposed method undergoes rigorous evaluation using diverse datasets, including both original sensor data and Conditional GAN-generated synthetic samples. Experimental results demonstrate substantial improvements across key performance metrics, including accuracy, precision, and sensitivity, highlighting the robustness and scalability of the approach for practical applications in WSNs.

This article is structured to provide a clear and comprehensive overview of the research. Section I outlines the research objectives and highlights their significance. Section II offers contextual background through a review of previous relevant studies. Section III details the methodology employed, including data acquisition and preparation. Section IV presents the main findings and evaluates model performance. Finally, Section V concludes by summarizing the study's outcomes and potential direction for future research in this field.

II. Related Work

Recent advances in IoT-based services reveal an upward trend in their application across various sectors. A significant factor in ensuring the quality of service from these IoT systems is the reliability of data transmitted by sensors. In real-world applications, sensors are generally deployed in harsh settings, such as extreme industrial or natural conditions, and are responsible for monitoring a range of physical parameters [22]. Low-cost sensors may experience abnormalities in their data over time as a result of aging, hardware failures, improper installation, or environmental impacts such as excessive temperature or humidity [23]. Many IoT applications rely on sensor data for decision-making and control, so unidentified faulty data can seriously affect system accuracy and reliability, potentially causing severe consequences [24].

Traditionally, fault detection and diagnosis in industrial and IoT settings faced significant challenges due to the complexity of physical systems. However, the emergence of ML has enabled data-driven approaches to overcome these limitations. ML techniques have been widely applied in various domains, including fault diagnosis of mechanical components in rotating machinery [25]. In WSNs, timely detection of sensor faults is critical for ensuring safety, improving data quality, prolonging network lifespan, and reducing response times [26], [27].

Traditional ML approaches have been extensively explored for sensor fault detection [28]. For instance, Miao et al. [29] proposed an online, distributed one-class support vector machine (SVM) for anomaly detection in WSNs, which demonstrated effectiveness in handling streaming data while reducing computational complexity. Similarly, Yoo et al. [30] employed an SVM-based approach for fault detection to assess system resilience under sensor degradation and faults. In another study, binary SVM classification was employed in [31] to identify normal and faulty signals, achieving high detection accuracy. While these methods show promise, they rely heavily on handcrafted features, limiting their scalability to diverse datasets.

To address these challenges, DL methods such as convolutional neural networks (CNNs) have emerged as powerful tools capable of directly extracting complex features from raw data. In [32], ensemble learning methods, including random forest (RF), stochastic gradient descent (SGD), and CNNs, were evaluated for multihop WSNs, demonstrating notable improvements in detecting common sensor faults such as offset, gain, and stuck-at errors. Furthermore, Saeed et al. [33] used the extra tree (ET) algorithm for diagnosing sensor faults, including hardover, drift, spike, erratic, data loss, stuck, and random faults in WSNs, which outperformed traditional techniques such as SVM and RF in accuracy and precision while offering reduced training time.

In a related study, Saeed et al. [33] also presented the development of a context-aware fault diagnostic system [34]. That technique efficiently detects and diagnoses the aforementioned sensor faults using an ET ensemble learning algorithm. The main feature was developing a two-stage fundamental component of an ML technique. Initially, a multilabel classification method is used to label faults based on their contextual cause, such as a hardware malfunction or a calibration error. The ET classifier is then trained using this context-enriched data, which significantly enhances diagnostic accuracy.

Advanced preprocessing techniques have also been explored to enhance the performance of DL models. Transformations, such as short-time Fourier transform (STFT) [35], GAF encoding [36], and spectrograms [37], [38], [39], have shown promise in effectively extracting meaningful features from raw data, enabling more accurate and robust model performance. For example, Gou et al. [40] employed scalograms as input to CNN models for fault detection in aeroengine control systems, while Zhao et al. [41] used modified CNN architecture that incorporated a Morlet wavelet basis function as the activation function in two of the intermediate convolutional layers, enhancing its ability to detect sensor faults in hydrogen gas sensors. Similarly, in [35], signals obtained from a drone were transformed into the time–frequency domain using STFT. The resulting signals were subsequently employed to train a four-layer CNN classifier. These methods highlight the importance of combining advanced preprocessing with DL models to improve fault classification.

Generative models, particularly GANs, have recently gained traction for their ability to handle data imbalance and generate synthetic fault data. In [36], a GAF-based GAN framework was introduced for fault detection, effectively leveraging temporal correlations in sensor data to distinguish between normal and faulty states. Similarly, Li et al. [42] used CNNs with GAF-encoded time-series data, achieving high accuracy and stability in fault classification. These approaches address key challenges, such as limited fault data and imbalanced datasets, further expanding the capabilities of fault detection systems.

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TABLE I
SUMMARY OF RELATED WORKS ON SENSOR FAULT DETECTION

Ref.	Approach	Datasets	Key Findings	Fault Types
[28]	SVM with statistical time-domain features.	Simulated WSNs Dataset.	Demonstrates high classification accuracy for sensor faults.	Bias, drift, precision degradation.
[29]	Distributed one-class SVM for anomaly detection.	Synthetic and real-world WSNs datasets.	Detects anomalies in streaming data with reduced computational complexity.	Binary (Normal vs abnormal).
[31]	SVM-based fault classification.	Simulated and real-world WSNs datasets.	Achieves high detection accuracy through feature selection techniques.	Binary (Normal vs Faulty).
[32]	Random Forest classifier for fault detection.	Real-world WSNs datasets with injected faults.	Outperforms classifiers like SVM and CNN in fault detection tasks.	Gain, offset, spike, data loss, stuck-at.
[33]	Extra Trees for sensor fault detection.	Simulated and real-world WSNs datasets.	Provides high detection accuracy with reduced computational requirements.	Bias, drift, spike, erratic, stuck.
[34]	CAFD scheme with Extra-Trees and ML techniques.	Humidity and temperature WSN sensor data.	Demonstrates high diagnostic accuracy for low-intensity faults.	Drift, hard-over, spike, erratic, stuck, data loss.
[35]	Deep learning and time-frequency analysis.	UAV sensor datasets with faults.	Demonstrates robust performance in UAV fault classification.	Bias, drift, abrupt faults.
[36]	WGAN-based digital twin model.	WSNs data with drift faults.	Enables early detection of drift faults with high accuracy.	Drift faults.
[40]	CNN with CWT for feature extraction and classification.	Simulated aero-engine control data.	Achieves high classification accuracy by integrating time-frequency features with CNN models.	Short circuit, open circuit, spike, bias, drift, periodic disturbance.
[41]	CNN with wavelet-based analysis for fault detection.	Iron oxide gas sensor data.	Effectively identifies multiple fault classes with high accuracy.	Multi-class.
[42]	CNN with GAF encoding for time-series data.	Machinery vibration datasets.	Achieves superior classification accuracy by transforming time-series data.	Inner race, outer race, ball faults.

A brief summary of related work, along with its key findings and classification limitations, is provided in Table I. In order to enhance anomaly detection in industrial robotic sensors, Lu et al. [43] developed enhanced GANs called MS-GANs, creating synthetic anomaly samples to boost detection accuracy for industrial robotic sensors. Experiments on sensor data demonstrated the effectiveness of the approach. Similarly, Lu et al. [44] introduced the multilabel time-series GANs (MTGANs), which generates synthetic electronic health record (EHR) data, particularly for rare disorders. This method effectively handles data imbalance by using a novel training technique along with a gated recurrent unit (GRU)-based generator to create realistic EHR sequences that are validated by various metrics. In another study [45], a Conditional GAN was proposed to develop a machining DT capable of generating time-series data representing vibration signals, thereby supporting the precise modeling of physical systems.

Despite the advancements in ML and DL methods, existing works often face limitations in generalizability and computational efficiency, particularly in resource-constrained WSN environments. Moreover, the integration of advanced preprocessing techniques like GAF with hybrid models remains underexplored. This study addresses these gaps by proposing a robust framework leveraging GAF encoding and hybrid DL models for enhanced fault detection in WSNs.

III. METHODOLOGY

The proposed framework for sensor fault detection in WSNs employs a hybrid dual-channel approach, integrating a CAE and Conditional GAN, to generate synthetic sensor readings that closely resemble real-world data employing Conditional GANs. This study addresses common WSN faults, including hardover, drift, spike, erratic, and stuck faults. Time-series sensor data are transformed into GAF images to preserve temporal relationships, and features are extracted using the CAE model. These features are then fed into a three-layer ANN for fault classification, as illustrated in Fig. 1. The simulation parameters of the WSN structure used in this study are summarized in Table II.

A. Data Acquisition

For ML and DL models to be effective, it is ideal to utilize real data collected under realistic scenarios. Specifically for fault diagnosis, it is essential to have a dataset containing both normal sensor readings and faulty sensor readings to enable comprehensive model training. However, there are no publicly available datasets that include faulty sensor readings. To address this gap, we introduced various common WSN sensor faults by generating synthetic faults in normal temperature sensor readings, an approach that has been utilized by several researchers before [33], [46], [47].

This research utilizes a publicly available dataset developed by the research teams at the University of North Carolina at Greensboro. Using TelosB motes, they gathered temperature and humidity readings to simulate both single-hop and multihop WSN situations [48]. TinyOS-based sensor nodes were used for data collection, capturing humidity and temperature readings in the immediate vicinity in both indoor and outdoor environments. In the single-hop scenario, two indoor and two outdoor sensors concurrently collected temperature and humidity readings and transmitted the data to a central base station.

The multihop scenario was intended to cover a larger area leading to a vast WSN setup, as illustrated in Fig. 2. Since each sensor node has a limited transmission range, intermediary routers relay the data from the nodes to the

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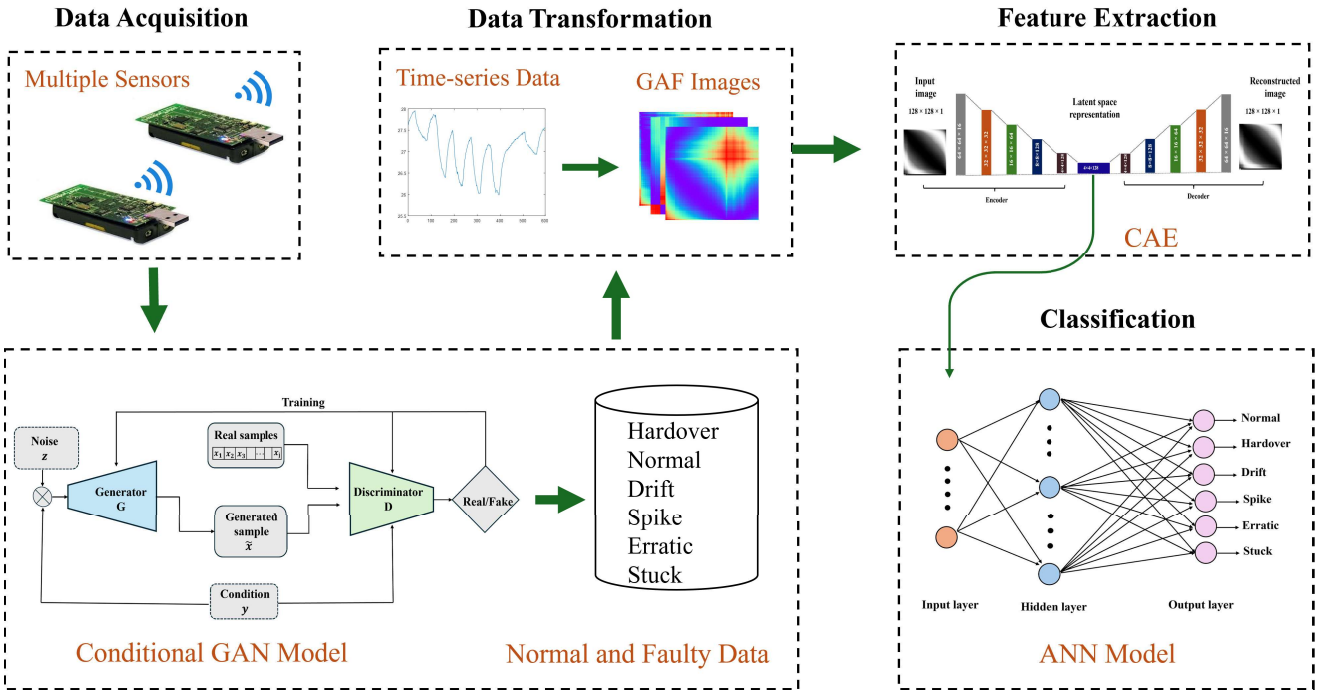


Fig. 1. Proposed workflow for sensor fault diagnosis in WSNs involves five stages: 1) data acquisition from multiple sensors; 2) data preparation using Conditional GANs to generate synthetic faulty and normal sensor data; 3) data transformation of time-series data into GAF images; 4) feature extraction via CAE; and 5) classification using an ANN to categorize sensor data into normal and various fault types.

TABLE II
SIMULATION PARAMETERS OF WSN STRUCTURE

Parameter	Value
Network Type	Multi-hop WSNs
Frequency Band	2.4 GHz (ISM Band)
Bandwidth	250 kbps
Data Rate	1 sample per 5 seconds
Number of Nodes	4 (2 Indoor, 2 Outdoor)
Packet Size	10 readings per packet
Sampling Duration	6 hours

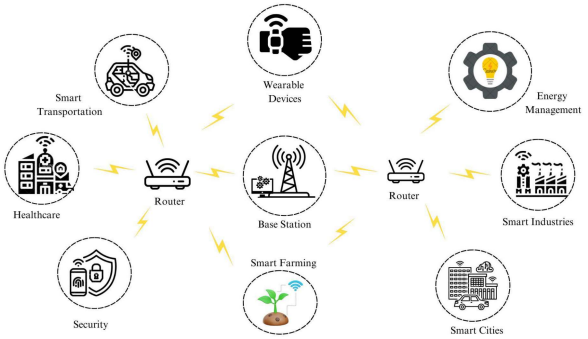


Fig. 2. WSN architecture for a multihop scenario.

base station. The multihop network made use of four sensors, and similar to the single-hop networks, four sensors were deployed, for gathering and transmitting data in the form of packets containing ten readings. To regulate the data flow, the data were then delivered to the base station. A sampling frequency of one reading every 5 s was used for data collected over a 6-h period.

During data collection, a hot water kettle was placed near one indoor and one outdoor sensor to deliberately alter the temperature and humidity measurements, leading to faults. However, in our research, these anomalous instances were carefully excluded from our analysis. Our study focuses on the multihop scenario, specifically considering the data from two indoor sensors. The proposed method is designed to detect and analyze faults at an intermediate router, allowing the base station to filter out the faulty data and preventing incorrect decisions based on it.

B. Data Preparation

This section outlines the data preparation techniques for effective sensor fault detection. It begins with Conditional GANs to generate synthetic data, enhancing diversity and addressing data scarcity. Next, synthetic fault injection replicates real-world sensor anomalies to evaluate fault detection models. Finally, time-series data are transformed into 2-D GAF representations, enabling enhanced feature extraction for fault classification. This sequential approach ensures a robust dataset, capturing diverse fault scenarios that are essential for evaluating diagnostic algorithms.

1) Conditional GANs: GANs are widely used for generating synthetic data that replicates the distribution of real-world datasets. However, traditional GANs lack control over specific data characteristics during the generation process and operate in an unsupervised manner. While they can generate realistic samples, they do not offer control over the class or type of data produced, making them unsuitable for purposes requiring targeted sample generation. To overcome this limitation, Mirza and Osindero [49] proposed Conditional GANs and expanding GAN capabilities from unsupervised to supervised learning. This extension enables the generation of samples conditioned

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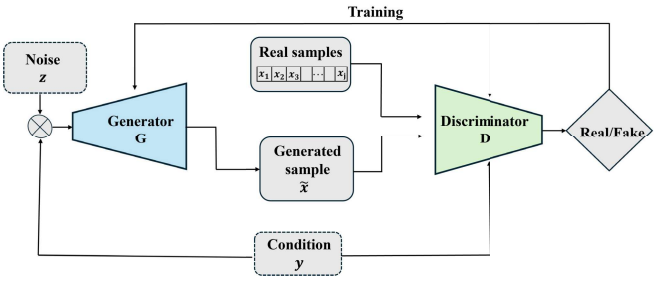


Fig. 3. Conditional GAN model architecture.

on specific input variables, allowing more controlled and focused data synthesis.

A typical GAN consists of two key components: a discriminator and a generator. The generator produces high-dimensional data samples from low-dimensional random input, typically represented as noise vectors. In contrast, the discriminator evaluates the authenticity of the generated samples, allocating a score between 0 and 1, where higher scores indicate greater similarity to real data. The generator and discriminator have an adversarial relationship because they are trained simultaneously with contradictory goals. The generator aims to create samples that the discriminator cannot distinguish from real data, while the discriminator attempts to improve its ability to differentiate between real and synthetic samples. This dynamic leads to continuous self-improvement in both components through iterative adjustments of their parameters. The optimization of a GAN is governed by the following objective function:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\big[\log D(x)\big] + \mathbb{E}_{z\sim p_{\text{data}}(z)} \times \big[\log(1 - D(G(z)))\big]. \quad (1)$$

Equation (1) represents the generated sample as $G(z)$, x represents the original sample, and z denotes the generator input of random noise.

In traditional GANs, only a random vector such as Gaussian noise can be input, meaning that the generated samples cannot be controlled. In contrast to basic GANs, Conditional GANs introduce a conditioning mechanism that guides the generation process. Fig. 3 shows the Conditional GAN structure. In this model, not only is the conditional variable y introduced into the generator, but it is also integrated into the discriminator.

Conditional variable y and a random noise vector are provided to generate samples. The discriminator evaluates whether the generated samples match conditional input y in addition to determining whether the samples are real. By preventing the generator from disregarding the conditions during training, this dual evaluation criterion ensures that the generated sample aligns with the given conditions. The conditional variable is incorporated into the optimization objective for the Conditional GANs, which is expressed as follows:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\big[\log D(x|y)\big] + \mathbb{E}_{z\sim p_{\text{data}}(z)} \times \big[\log(1 - D(G(z|y)))\big]. \quad (2)$$

2) Synthetic Fault Injection: Synthetic fault injection into time-series data is crucial for evaluating the robustness and accuracy of fault detection systems. After generating synthetic samples, five common fault types in WSNs, including hardover, drift, spike, erratic, and stuck faults, are simulated to test diagnostic algorithms. This approach supports early fault detection and impact mitigation by incorporating multiple fault intensities to reflect varying severities. Fault rates are systematically defined at 0.2, 0.4, 0.6, 0.8, and 1.0, representing incremental severity levels. Each fault type is dynamically scaled within this range, enabling comprehensive performance evaluation across diverse scenarios. Fig. 4 provides visual examples of these faults, which are related to either sensor performance issues or inaccurate data collection. The sensor node data, represented as a time series, can be mathematically expressed as

$$d(n, t, f(t))). \quad (3)$$

Fault types and their associated synthetic injection methods are described as follows.

a) Hardover fault: A hardover fault occurs when the output of a sensor consistently deviates from its normal state by a fixed higher value. This type of fault is characterized as a constant bias added to the sensor output. Common causes include mechanical displacement, gradual sensor drift, or improper calibration during installation. Mathematically, a hardover fault can be expressed as

$$S_n^{\text{hardover}} = S_n^{\text{normal}} + S \cdot b, \quad b = \text{constt} \quad (4)$$

where S_n^{normal} denotes the nonfaulty signal, b represents the bias magnitude, and S adjusts the fault intensity between 0.2 and 1.0. The fault intensities range from 0.2 (low bias) to 1.0 (high bias), corresponding to deviations between 2 and 10 units.

b) Drift fault: A drift fault is a gradual, linear increase in the sensor output over time, starting from its normal state. This fault is generated by adding a bias term that increases linearly with each data point. Mathematically, the drift fault signal, denoted as S_n^{drift}, for the nth data point, is modeled as

$$S_n^{\text{drift}} = S_n^{\text{normal}} + S \cdot (n \times b_0) \quad (5)$$

where S_n^{normal} represents the original nonfaulty signal; b_0 is the initial drift magnitude, dynamically scaled by S, simulating intensities from 0.2 to 1.0; and n indicates the data point index. Fault intensities vary from 0.2 to 1.0, corresponding to drift rates between 0.002 and 0.01 units per time step. This approach effectively models gradual deterioration, ranging from subtle changes to significant deviations, to evaluate fault detection robustness under progressive degradation.

c) Spike fault: Spike faults are abrupt, high-amplitude deviations in the sensor output, characterized by large, intermittent biases added to specific elements of the nonfaulty signal. These faults are highly disruptive and often arise due to electromagnetic interference, sudden shocks, or data transmission errors. Mathematically, a spike fault can be represented as

$$s_N^{\text{spike}} = s_N^{\text{normal}} + S \cdot b_N$$
$$b_N = \begin{cases} b, & \text{if } N = \nu \times \eta \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

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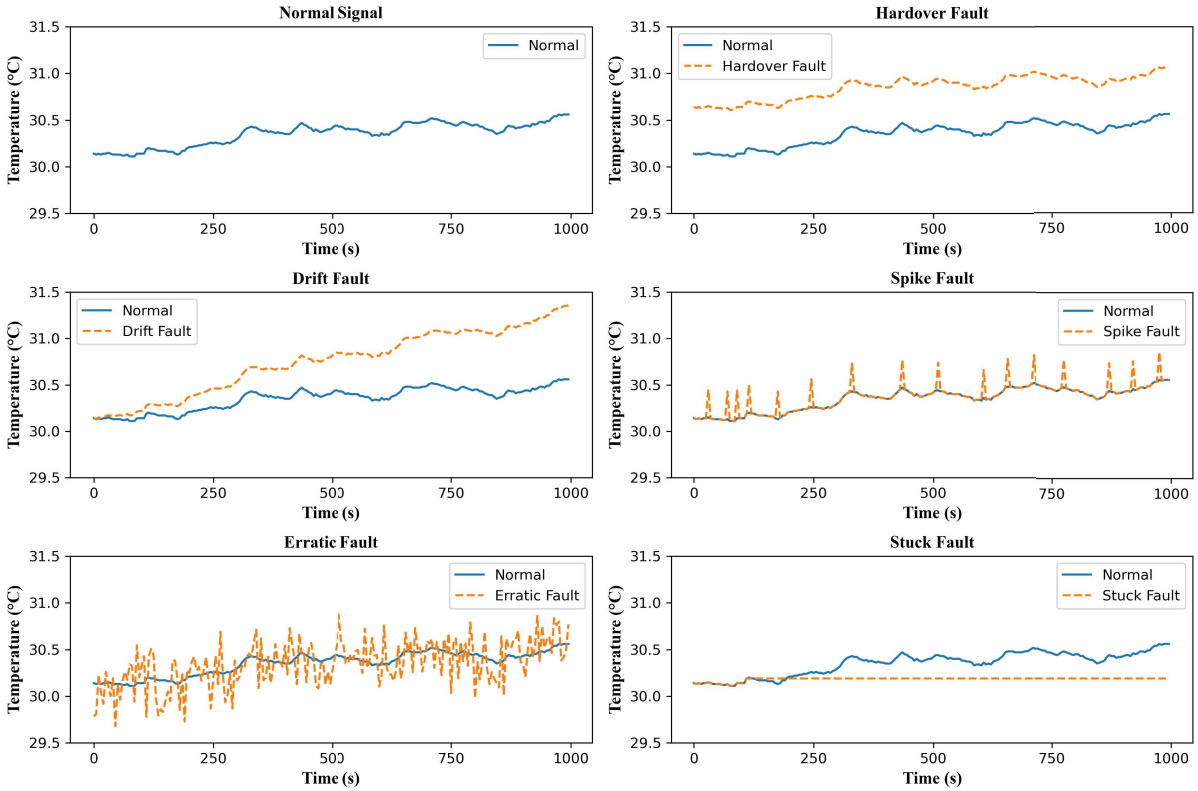


Fig. 4. Sample plots of faulty signals compared to a normal signal.

where S_N^{spike} represents the nonfaulty signal and b_N denotes the spike amplitude, dynamically scaled by S, reflecting intensities from 0.2 to 1.0. ν represents a sequence of natural numbers ($\nu = 1, 2, 3, \ldots$), and η is the intensity scaling factor, constrained as $\eta \geq 2$. Fault intensities range from 0.2 to 1.0, generating spikes of varying amplitudes to evaluate the performance of fault detection systems under sudden disturbances.

d) Erratic fault: Erratic faults exhibit a significant increase in the variance of sensor output, far exceeding normal levels. These faults emulate disturbances caused by environmental fluctuations, hardware defects, or unstable connections. Mathematically, an erratic fault can be modeled as

$$s_n^{\text{erratic}} = s_n^{\text{normal}} + \dot{S}_n, \quad \dot{S}_n \sim \mathcal{N}(0, \dot{\delta}^2) \tag{7}$$

where S_n^{normal} represents the nonfaulty signal and $\dot{S}_n$ is a high-variance signal with a mean of 0. The variance of $\dot{S}_n$ satisfies S, which dynamically scales the variance of the fault, controlling noise levels between 0.1 and 0.5 units. Noise variances range from 0.1 to 0.5 units, reflecting transitions from mild to severe random fluctuations that can challenge system stability. Fault intensities are scaled from 0.2 (low noise) to 1.0 (high noise) to evaluate detection robustness under increasing randomness.

e) Stuck fault: A stuck fault can be either temporary or persistent and is characterized by minimal or no variation in the sensor output over a specific period of time. In cases of total failure, the sensor output remains constant. In this work, we introduce five different intensities by varying the duration of stuck periods. Mathematically, a stuck fault is represented as

$$S_n^{\text{stuck}} = \alpha, \quad \alpha = \text{const} \tag{8}$$

where S_n^{stuck} denotes the stuck fault and α is applied uniformly across all indices of the nonfaulty signal. Fault intensities range from 0.2 (short stuck periods) to 1.0 (extended stuck periods).

3) GAF Imaging: The GAF concept was first presented by Wang and Oates [50] as a way to transform 1-D data into a 2-D representation. The primary motivation for using GAF images in fault diagnosis stems from their compatibility with convolutional models, which is commonly utilized by DL models, for processing 2-D imaging data. The model typically consists of layers such as pooling and convolutional layers. These visual representations can be utilized more efficiently if 1-D data are transformed into 2-D visuals. Using this approach allows a CNN-based model to fully leverage its strengths in feature extraction and provides classification that processes data efficiently. Moreover, there is a temporal correlation in the track circuit's 1-D data. This method allows 1-D data to be processed into 1-D images while preserving a temporal correlation in the images, which allows the temporal correlation within time-series data to be captured. In addition, no significant information is lost during the data transformation process.

By transforming sensor readings into GAF images, a time series with n samples or observations, $X_n = (x_1, x_2, x_3, \ldots, x_n)$, is rescaled to start the encoding

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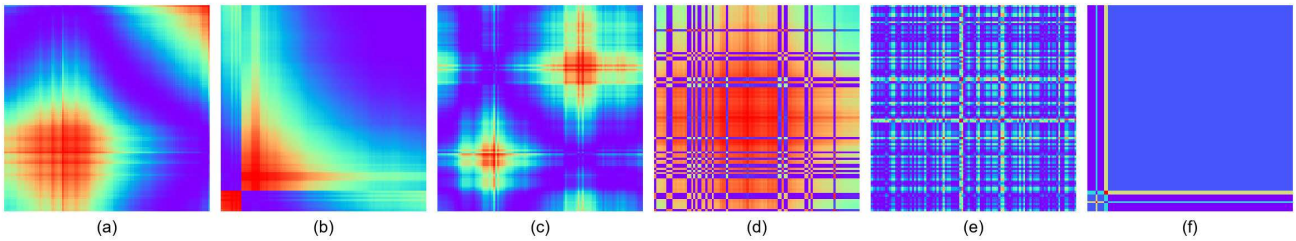


Fig. 5. GAF images corresponding to different fault scenarios in the dataset. (a) Normal data. (b) Hardover fault. (c) Drift fault. (d) Spike fault. (e) Erratic fault. (f) Stuck fault.

process

$$\tilde{x}_i = \frac{x_i - \min(X_n)}{\max(X_n) - \min(X_n)}. \tag{9}$$

We transform the scaled time series into polar coordinates. The angular cosine of the time-dependent value is used to indicate the angle, while the radius serves as the timestamp

$$\begin{cases} \phi_i = \arccos(\tilde{x}_i) \\ r_i = \dfrac{t_i}{n}. \end{cases} \tag{10}$$

Given that the value of x_i falls within the range -1 to 1, the corresponding angle, ϕ_i, lies between 0 and π. Radius r_i in the polar coordinate system, and the timestamp is normalized by dividing it by the length of the time series n. Wang and Oates [50] state that this transformation is bijective, guaranteeing the preservation of temporal relationships. The resulting angular values, ϕ_i, are subsequently used to generate a Gramian-like matrix, as described in the following:

$$\text{GAF} = \begin{pmatrix} \cos(\phi_1+\phi_1) & \cos(\phi_1+\phi_2) & \cdots & \cos(\phi_1+\phi_n) \\ \cos(\phi_2+\phi_1) & \cos(\phi_2+\phi_2) & \cdots & \cos(\phi_2+\phi_n) \\ \vdots & \vdots & \ddots & \vdots \\ \cos(\phi_n+\phi_1) & \cos(\phi_n+\phi_2) & \cdots & \cos(\phi_n+\phi_n) \end{pmatrix} \tag{11}$$

$$\text{GAF} = \tilde{X}' \cdot \tilde{X} - \sqrt{I - \tilde{X}^2}' \cdot \sqrt{I - \tilde{X}^2} \tag{12}$$

where I is a unit row vector of length n and $\tilde{X}'$ is the transpose of the rescaled time series. Fig. 5 displays the samples of GAF-encoded images from a normal signal as well as from various sensor faults.

C. Feature Extraction and Classification

1) CAE for Feature Extraction: The CAE can be an effective tool in the field of image processing by learning hierarchical data representations. These particular neural networks, first introduced by Ballé et al. [51] and Theis et al. [52], can be designed to encode and decode data with spatial hierarchies, such as images. Convolutional layers, from which CAE effectively captures spatial information, make them especially useful at image processing tasks. Approaching the identity function while working within specific parameters (such as the maximum number of neurons in hidden layers) is the main objective of a CAE. By compressing input, $x \in \mathbb{R}^n$, the encoder portion of a CAE operates as a funnel and makes n compressed feature maps, $x \in \mathbb{R}^{n \times l \times l}$, in which $l \times l$ pixels cover each feature map, comprising the input. The convolutional kernel has dimensions $d \times d$ where $d \leq 1$.

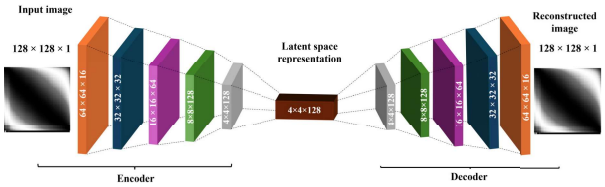


Fig. 6. CAE model architecture.

Initially, the input image goes through an encoding process that divides it into small patches ($d \times d$ size pixels) denoted as x_i where $i = 1, 2, 3, \ldots, p$. For each patch, w_j, where the jth convolution kernel is utilized for processing convolution operations. This results in the computation of the neuron value o_{ij} for $j = 1, 2, 3, \ldots, m$ in the output layer

$$o_{ij} = f(x_i) = \sigma\left(w_j \cdot x_i + b\right) \tag{13}$$

where the nonlinear activation function is denoted by σ. The rectified linear unit (ReLU) activation function is used in this context

$$\text{ReLU}(x) = \begin{cases} x, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0. \end{cases} \tag{14}$$

After the convolutional decoding stage, the output o_{ij} undergoes encoding, and x_i is reconstructed using o_{ij} to produce $\hat{x}_i$

$$x_i = f'(o_{ij}) = \varphi\left(w_i \cdot o_{ij} + \hat{b}\right). \tag{15}$$

SGD is utilized within the CAE layer to iteratively refine weights and minimize errors, leading to the optimization of model parameters to generate feature maps. $\hat{x}_i$ is reconstructed for each instance after the convolutional encoding and decoding process. During the reconstruction phase, patches p of size $d \times d$ are evaluated, and the mean square error between original input patch x_i and reconstructed patch $\hat{x}_i$ is calculated, where $i = 1, 2, 3, \ldots, p$. The reconstruction error is quantified by a cost function, as defined in the following equations:

$$JC(\theta) = \frac{1}{p} \sum_{i=1}^{p} L\left[x_i, \hat{x}_i\right] \tag{16}$$

$$LC\left[x_i, \hat{x}_i\right] = \|x_i - \hat{x}_i\|^2 = \|x_i - \varphi(\sigma(x_i))\|^2. \tag{17}$$

Backpropagation and optimization techniques are employed in the study to reduce the loss, particularly while training the CAE model with several layers of encoders and decoders.

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TABLE III
PARAMETERS USED IN SIMULATING THE CAE

Encoder		
Layer Type	**Output Shape**	**Parameters**
Conv2D	64 × 64 × 16	160
Conv2D	32 × 32 × 32	4640
Conv2D	16 × 16 × 64	18496
Conv2D	8 × 8 × 128	73856
Conv2D	4 × 4 × 128	147584
Decoder		
Layer Type	**Output Shape**	**Parameters**
Dense	2048	526336
Conv2DTranspose	8 × 8 × 128	147584
Conv2DTranspose	16 × 16 × 64	73792
Conv2DTranspose	32 × 32 × 32	18464
Conv2DTranspose	64 × 64 × 16	4624
Conv2DTranspose	128 × 128 × 1	145
Trainable Parameters: 1540225		
Non-trainable Parameters: 0		
Total Parameters: 1540225		

Initially, the GAF images are fed into the model and subsequently pass through a series of convolutional layers. Table III presents the architecture of the CAE and details the input and output dimensions of each layer. High peak signal-to-noise ratio values for GAF images of signals, which are 58.43 dB, indicate the effectiveness of the CAE. These metrics demonstrate how accurate the image reconstruction process is. The layer-wise architecture of the implemented CAE is illustrated in Fig. 6.

2) ANNs for Classification: ANNs, which are constructed to represent the complex relationships between input and outputs, have been widely utilized across various fields. A typical feedforward ANN is composed of an input layer, one or more hidden layers, and an output layer, In this study, we employ a three-layer feedforward neural network to implement the proposed method for identification and classification of various sensor faults in WSNs.

The three-layer feedforward neural network used in this study introduces an input vector, $x_n = [x_1, x_2, \ldots, x_n]$, where $n = 256$ represents the number of input elements and the number of output neurons corresponding to six fault classes. To ensure stability and high convergence rates, we experimented with the number of hidden layer neurons through trial and error. Networks with eight or more hidden neurons demonstrated adequate performance for this application. Although this configuration may not be fully optimized, it was selected on trial and error based to meet predefined training error criteria (the issue of optimization is beyond the scope of this article). During training, biases and connection weights are iteratively adjusted to minimize the error between predicted and actual outputs. Training continues until the average error across the dataset falls below a predefined threshold.

IV. RESULTS AND DISCUSSION

In this section, we present a comprehensive evaluation of the proposed method, emphasizing its performance metrics and comparisons with current state-of-the-art models. We provide an in-depth review of the experimental setup, dataset descriptions, and performance assessments, along with a detailed analysis of the results. Moreover, we discuss how data augmentation, using Conditional GANs, optimized hyperparameter selection, and computational efficiency are evaluated to validate the proposed model's effectiveness.

TABLE IV
SPECIFICATIONS OF THE COMPUTING ENVIRONMENT

Component	Specification
Processor	AMD Ryzen 9 5900X, 12-Core, 3.7 GHz
RAM	64 GB
GPU	NVIDIA GeForce RTX 3080
Operating System	Windows 11 Pro
Python Version	3.9.18
TensorFlow Version	2.10.1
Keras Library	TensorFlow Backend
CUDA Version	11.2

TABLE V
COMPARISON OF THE ORIGINAL DATASET AND TWO SYNTHETIC DATASETS FOR EACH CLASS. THE SYNTHETIC DATASETS SIGNIFICANTLY INCREASED THE SAMPLE SIZE FOR EACH CLASS

Class	Original Dataset	Synthetic Dataset (Lower Samples)	Synthetic Dataset (Higher Samples)
Normal	92	649	1206
Hardover	92	649	1206
Drift	92	649	1206
Spike	92	649	1206
Erratic	92	649	1206
Stuck	92	649	1206
Total	**552**	**3894**	**7236**

A. Experimental Setup

A series of experiments was conducted to evaluate the effectiveness of the proposed models for detecting sensor faults through DL techniques. These simulations were executed in a high-performance computing environment, as outlined in Table IV. The computational setup featured an Advanced Micro Devices (AMD) Ryzen 9 5900X 12-core processor, 64-GB RAM, and an NVIDIA GeForce Ray Tracing Texel eXtreme (RTX) 3080 graphics processing unit (GPU), ensuring efficient data processing and enabling accelerated model training and evaluation. The GPU's computational capabilities proved instrumental in handling large datasets and executing complex simulations, particularly during the training phases of the CAE and ANN frameworks.

All model development, training, and evaluation were implemented in Python version 3.9.18, leveraging TensorFlow version 2.10.1 along with the Keras API to build and refine DL architectures. GPU acceleration, powered by the Compute Unified Device Architecture (CUDA) toolkit and cuDNN, significantly boosted computational speed and enabled parallel processing. For data handling and visualization, NumPy 1.23.5 and Matplotlib were employed, while Scikit-learn 1.5.1 provided classification metrics. Supported by an NVIDIA RTX 3080 GPU and high-performance hardware, this software

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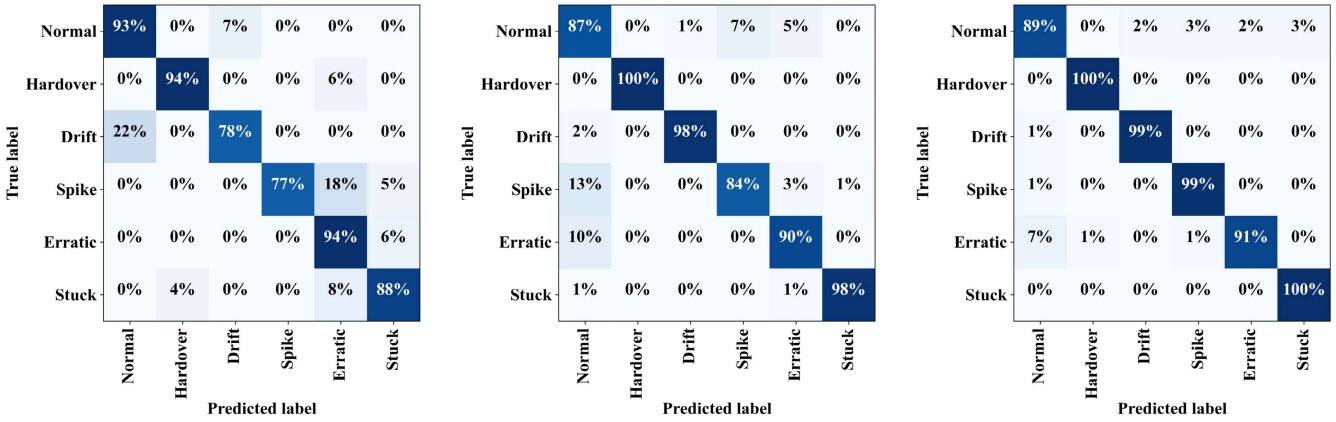


Fig. 7. Confusion matrices from sensor fault diagnosis in a WSNs using (a) original dataset, (b) synthetic dataset with a lower number of samples, and (c) synthetic dataset with a higher number of samples.

configuration efficiently handled large datasets, allowing the model to be trained using both synthetic and original data.

B. Experimental Results

Three distinct datasets were employed for evaluation: the original dataset, a lower synthetic dataset (LSD), and a higher synthetic dataset (HSD). These dataset collections were designed to assess the model's generalization capabilities by incorporating data augmented using Conditional GANs. Table V presents the sample size comparisons for each dataset.

The datasets were partitioned into training and testing sets, with an 80%–20% split. The generated synthetic datasets were used during the training phase, while the original sensor data were reserved for testing purposes. To evaluate the performance, several metrics were computed: accuracy, $F1$-score, precision, sensitivity, and confusion matrices. These metrics were selected to provide a comprehensive assessment of the model's effectiveness and its ability to identify and differentiate between various sensor fault conditions. The confusion matrix was utilized to further analyze classification accuracy, allowing for a detailed comparison between predicted and actual labels of the test samples. True positives (TPs) denote instances correctly identified as faulty, while true negatives (TNs) refer to samples accurately classified as normal. Conversely, false positives (FPs) are instances wrongly labeled as faulty, and false negatives (FNs) are instances incorrectly identified as normal

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{18}$$

$$F1\text{-score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{19}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{21}$$

To rigorously validate the reliability and robustness of the proposed model, several performance metrics were initially computed using the original dataset, yielding an average accuracy of 84.68%, an $F1$-score of 83.41%, a precision of 84.67%, and a sensitivity of 84.67%. However, the confusion matrix, shown in Fig. 7(a), revealed a relatively high rate of misclassifications, particularly for rare fault types. This observation highlights limitations in both generalizability and robustness when relying solely on the original dataset.

To address these issues and confirm the reliability of the model, two additional datasets containing synthetic fault signals were generated via Conditional GANs: the LSD and the HSD. With the LSD, model performance demonstrated significant improvement, achieving an accuracy of 91.53%, an $F1$-score of 91.12%, a precision of 91.12%, and a sensitivity of 91.16%. As illustrated in Fig. 7(b), the results highlight a reduction in misclassifications attributed to the additional synthetic data, which increased both sample diversity and training stability. Further enhancements were observed with the HSD, where the model attained an accuracy of 95.93%, an $F1$-score of 95.84%, a precision of 95.87%, and a sensitivity of 95.88%. The confusion matrix in Fig. 7(c) emphasizes the model's robustness in handling diverse and realistic synthetic samples, minimizing misclassifications and outperforming both the original dataset and the LSD.

Furthermore, 95% confidence intervals (CIs) for each performance metric were computed using the normal approximation method, $\text{CI} = P \pm 1.96 \cdot (p(1-p)/n)^{1/2}$, where p represents the metric value and n is the sample size. These intervals define the statistical range within which the true metric values are likely to fall with 95% confidence. Notably, the LSD and HSD exhibited narrower CIs than the original dataset as shown in Table VI, highlighting enhanced consistency and diminished variability, especially for infrequent fault classes. Consequently, incorporating synthetic data generated by Conditional GANs significantly improved model performance and generalizability, affirming the usefulness of data augmentation in fault detection for WSNs.

A comparative analysis of the ANN's performance across the different datasets is illustrated in Fig. 8, demonstrating the effectiveness of enhancing model performance by using synthetic data. The results, summarized in Table VII, highlight

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TABLE VI
COMPARATIVE PERFORMANCE METRICS WITH CIS FOR ORIGINAL DATASET, LSD, AND HSD IN A MULTICLASS CLASSIFICATION PROBLEM

Dataset	Metric	Normal (%)	Hardover (%)	Drift (%)	Spike (%)	Erratic (%)	Stuck (%)
Original Dataset	Accuracy	93.0 ± 2.1	88.0 ± 2.4	50.0 ± 3.1	77.0 ± 2.9	94.0 ± 1.7	92.0 ± 2.2
	F1-Score	72.0 ± 2.3	88.0 ± 2.5	64.0 ± 2.7	85.0 ± 2.8	81.0 ± 2.5	94.0 ± 2.3
	Precision	60.0 ± 2.4	88.0 ± 2.6	90.0 ± 2.8	95.0 ± 2.9	72.0 ± 2.6	96.0 ± 2.4
	Sensitivity	93.0 ± 2.1	88.0 ± 2.4	50.0 ± 3.1	77.0 ± 2.9	93.0 ± 1.7	92.0 ± 2.2
Lower Synthetic Dataset	Accuracy	71.0 ± 1.4	99.0 ± 0.7	98.0 ± 0.9	89.0 ± 1.6	90.0 ± 1.3	99.0 ± 0.7
	F1-Score	74.0 ± 1.6	99.0 ± 0.7	98.0 ± 0.9	88.0 ± 1.5	88.0 ± 1.4	99.0 ± 0.7
	Precision	76.0 ± 1.3	99.0 ± 0.7	98.0 ± 0.9	87.0 ± 1.7	89.0 ± 1.4	99.0 ± 0.7
	Sensitivity	71.0 ± 1.4	99.0 ± 0.7	98.0 ± 0.9	90.0 ± 1.6	90.0 ± 1.3	99.0 ± 0.7
Higher Synthetic Dataset	Accuracy	87.0 ± 0.8	99.0 ± 0.5	99.0 ± 0.7	99.0 ± 0.6	91.0 ± 0.9	99.0 ± 0.6
	F1-Score	89.0 ± 0.9	99.0 ± 0.5	98.0 ± 0.8	97.0 ± 0.8	94.0 ± 0.8	99.0 ± 0.5
	Precision	91.0 ± 0.8	99.0 ± 0.5	98.0 ± 0.8	95.0 ± 0.7	96.0 ± 0.7	98.0 ± 0.6
	Sensitivity	87.0 ± 0.8	99.0 ± 0.5	99.0 ± 0.7	99.0 ± 0.6	91.0 ± 0.9	99.0 ± 0.6

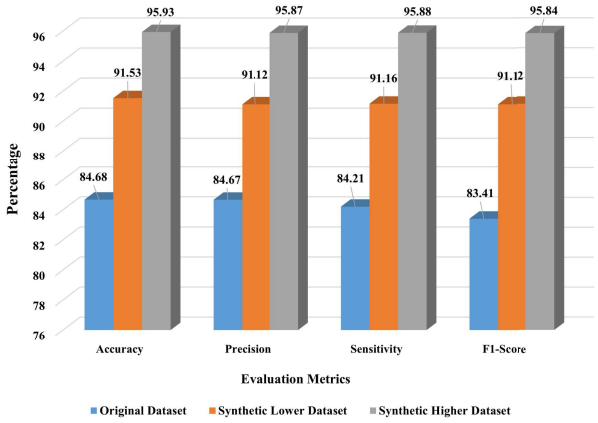


Fig. 8. Comparison of evaluation metrics for the original dataset, lower synthetic samples, and higher synthetic samples.

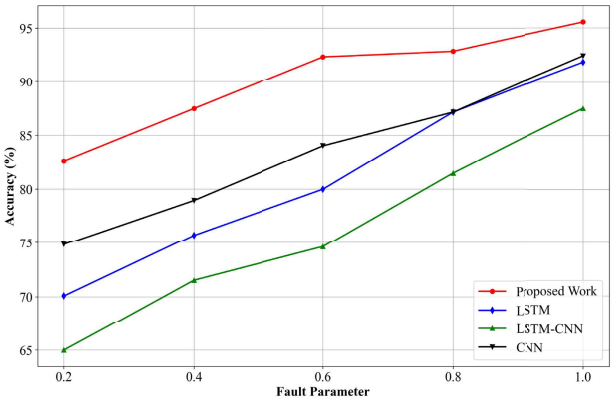


Fig. 9. Accuracy comparison of the proposed model with LSTM, LSTM-CNN, and CNN across different fault parameters.

TABLE VII
COMPARATIVE EVALUATION OF MODEL PERFORMANCE ACROSS DIFFERENT SENSOR DATASETS

Dataset	Accuracy (%)	F1-Score (%)	Precision (%)	Sensitivity (%)
Original	84.68	83.41	84.67	84.21
LSD	91.53	91.12	91.12	91.16
HSD	95.93	95.84	95.87	95.88

the significant impact of synthetic data on classification performance; when using the original dataset, the model achieved an average accuracy of 84.68%, while the dataset with more synthetic samples yielded an accuracy of 91.53%, and the dataset with even more synthetic samples achieved 95.93% accuracy (an 11.25% improvement). Similarly, the $F1$-score, precision, and sensitivity improved by 12.43%, 11.2%, and 11.67%, respectively.

To evaluate the model's robustness across different fault magnitudes, experiments were conducted at low, medium, and high severity levels, corresponding to fault rates of 0.2, 0.4, 0.6, 0.8, and 1.0 within the datasets. For instance, a fault parameter of 0.2 indicates the lowest fault severity, whereas 1.0 represents the highest. As depicted in Fig. 9, the average classification accuracy of the proposed model was compared against CNN, long short-term memory (LSTM), and LSTM-CNN architectures across six distinct classes, i.e., normal samples, and five fault types (hardover, drift, spike, erratic, and stuck) each evaluated at different fault severities (0.2, 0.4, 0.6, 0.8, and 1.0).

The proposed approach, leveraging GAF images incorporating CAE and ANN, demonstrated consistently superior performance, achieving up to 95% accuracy even under the most severe fault conditions. Its spatial encoding technique effectively captures temporal dependencies, while the CAE enhances feature extraction and suppresses noise, enabling the ANN classifier to maintain high accuracy across all fault levels. In contrast, models trained directly on raw time-series data, such as LSTM and LSTM-CNN, exhibited difficulties in handling complex and noisy patterns, particularly as fault severity increased. Their dependence on sequential memory cells makes them susceptible to overfitting and limits their ability to detect localized variations induced by faults. Although the CNN model, which also utilized GAF images, outperformed time-series-based models, it lagged behind the proposed approach due to its deeper architecture, leading to higher computational costs. The comparison of accuracy and training time for different models across fault severities is presented in Table VIII.

Fig. 10 presents the training time required for the models under varying fault severity levels. The proposed approach consistently demonstrates the shortest training duration,

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TABLE VIII
IMPACT OF FAULT SEVERITY ON MODEL ACCURACY AND TRAINING TIME FOR SENSOR FAULT DETECTION

Fault Severity	Proposed Work		LSTM		LSTM-CNN		CNN	
	Acc (%)	Time (sec)	Acc (%)	Time (sec)	Acc (%)	Time (sec)	Acc (%)	Time (sec)
0.2	82.7	95.15	70.1	231.02	65.1	174.91	74.9	129.03
0.4	87.5	95.50	75.8	238.56	71.6	176.75	78.9	130.39
0.6	92.4	96.79	79.9	232.70	74.7	171.31	84.1	126.83
0.8	92.9	98.50	87.2	234.95	81.5	175.31	87.1	129.19
1.0	95.6	100.07	91.9	239.44	87.6	185.91	92.3	132.19

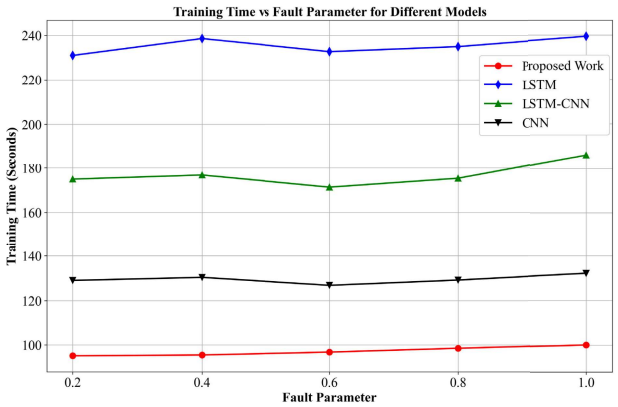


Fig. 10. Comparison of computational training time for the proposed model, LSTM, LSTM-CNN, and CNN across different fault parameters.

remaining under 105 s across all fault parameters. This efficiency is attributed to the CAE's ability to extract meaningful and rich features while reducing dimensionality, integrated with the lightweight design of the ANN classifier. These attributes make the proposed method ideal for real-time applications and deployment in resource-constrained environments.

In contrast, the LSTM model exhibits the longest training time, averaging approximately 235 s, primarily due to its sequential processing nature and reliance on memory cells. Although the LSTM-CNN model is slightly faster than LSTM, it still lags behind image-based methods due to the complexity of its hybrid architecture. The CNN model, which also processes GAF images, performs better than time-series-based models but requires greater computational resources than the proposed method, attributable to its deeper network layers. These findings underscore the superior efficiency and scalability of the proposed framework compared to alternative approaches. The hyperparameters used for training the classifiers were carefully fine-tuned, and their configurations are summarized in Table IX.

To further support the robustness of the proposed work, additional evaluations were conducted using two different datasets. The first dataset, VBL-VA001, is a publicly available laboratory-scale vibration analysis (VA) dataset developed by Vibrastic Lab (VBL) at the Department of Engineering Physics, Sepuluh Nopember Institute of Technology, Surabaya, Indonesia [53]. It comprises vibration signals captured from rotating machinery operating under diverse conditions. The second dataset is the WSN Dataset, which includes both temperature and humidity data. For this evaluation, data from both

TABLE IX
HYPERPARAMETERS AND CONFIGURATIONS USED FOR TRAINING DIFFERENT MODELS, INCLUDING LSTM, LSTM-CNN, CNN, CAE + ANN, AND CONDITIONAL GANs

Classifiers	Parameters
LSTM	optimizer = adam learning_rate = 0.001 batch_size = 32 epochs = 200 dropout = 0.3 loss = categorical_crossentropy
LSTM-CNN	optimizer = adam learning_rate = 0.0005 batch_size = 32 epochs = 200 dropout = 0.25 loss = categorical_crossentropy
CNN	optimizer = adam learning_rate = 0.001 batch_size = 32 epochs = 200 dropout = 0.25, 0.5 loss = sparse_categorical_crossentropy
CAE+ANN	optimizer = adam learning_rate = 0.001 batch_size = 32 epochs = 100 dropout = 0.2 loss = categorical_crossentropy
Conditional GANs	• **Generator:** hidden_layers = 128 → 256 → 512 → 1 activation_function = ReLU output_activation = Linear latent_dim = 100 dropout = 0.5 • **Discriminator:** hidden_layers = 512 → 256 → 128 → 1 activation_function = ReLU, Sigmoid (output) dropout = 0.5 • **Training Parameters:** optimizer = adam learning_rate = 0.001 batch_size = 64 epochs = 10,000 loss = binary_crossentropy • **Conditional Input:** time_steps (scaled) = Min-Max (0, 1)

the temperature and humidity sensors were utilized. Faults were injected into these datasets using the approach described in Section III, as no publicly available datasets with preexisting sensor fault data were available. The time-series data were then transformed into image representations. The models were trained using the original dataset size, and performance metrics for each dataset are summarized in Table X. The results presented in the table demonstrate that the proposed model

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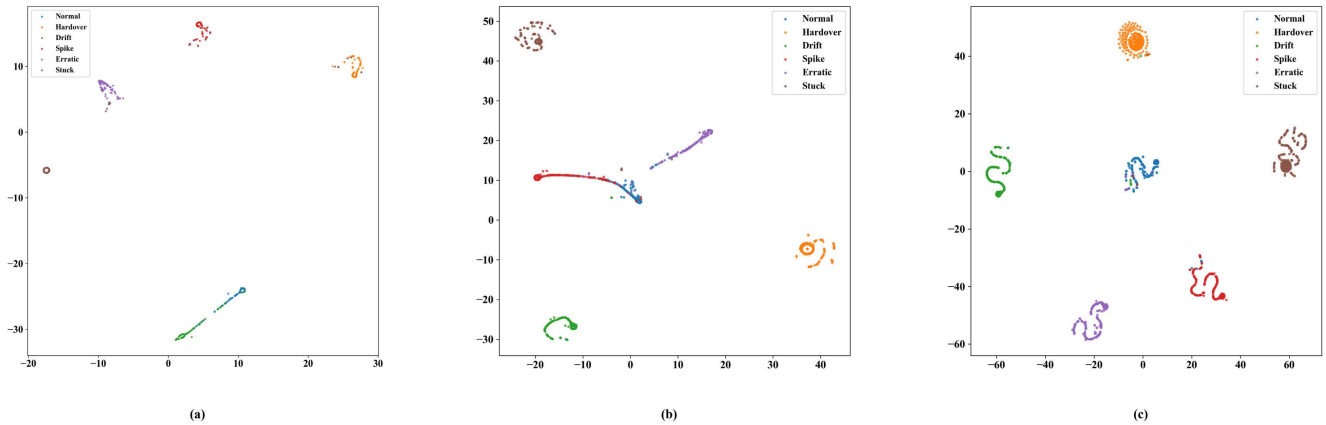


Fig. 11. t-SNE visualizations of sensor faults in WSNs across three datasets. (a) Original. (b) Lower synthetic samples. (c) Higher synthetic samples. Increasing dataset volume improves cluster separation, enhances feature representation, and boosts classification performance, demonstrating the effectiveness of larger synthetic datasets for fault differentiation in WSNs.

TABLE X
PERFORMANCE METRICS FOR DIFFERENT SENSOR DATASETS

Dataset	Accuracy	F1-Score	Precision	Sensitivity
WSNs Temperature	95.93 %	95.84 %	95.87 %	95.88 %
WSNs Humidity	92.71 %	92.01 %	91.94 %	92.88 %
VBL Vibration	91.70 %	91.55 %	91.61 %	91.68 %

delivers consistent and reliable performance across multiple datasets, reinforcing its adaptability and generalizability for fault detection tasks.

The performance of the proposed model was further compared against existing approaches using the same dataset. Saeed et al. [33] employed an extremely randomized tree algorithm for sensor fault classification, achieving an accuracy of 81.20% and an average $F1$-score of 81.92% for multiclass sensor fault detection. Similarly, Saeed et al. [34] reported an $F1$-score of 85.42% for multiclass fault classification, which, while competitive, remains lower than the 95.84% $F1$-score achieved by our model. Furthermore, a study [36] focused on binary classification between normal and drift faults, reporting average accuracy, precision, recall, and $F1$-score values of 98.7%, 97.5%, 99.9%, and 98.7%, respectively. In comparison, our model achieved comparable results for drift fault detection, with accuracy, precision, recall, and $F1$-score values of 99.12%, 97.41%, 99.12%, and 98.26%, respectively. Notably, our model achieved these results while addressing multiclass problems, which further underscores the efficiency, scalability, and robustness of the proposed framework, making it a more versatile solution for sensor fault detection in complex and diverse environments.

To visually evaluate the performance of our proposed model across different dataset volumes, we utilized t-SNE plots for a comparative analysis of the feature spaces. These plots effectively illustrate the clustering patterns of temperature sensor data points, with each cluster representing different conditions. The t-SNE visualizations for each dataset provide insights into how features are distributed and separated within a complex, multidimensional space. The model barely distinguished the various conditions in the original dataset. In contrast, with synthetically generated samples, the model not only distinguished normal and faulty samples but accurately classified the different types of fault in both datasets. Fig. 11 illustrates the t-SNE plots, emphasizing the enhanced classification capabilities of a model trained on a dataset enriched with synthetic samples compared to using the original dataset and using the dataset with even more synthetic samples.

V. CONCLUSION AND FUTURE WORK

This study proposed an innovative hybrid approach for sensor fault detection in WSNs, addressing critical challenges such as data scarcity and the complexity of identifying diverse fault types in IoT environments. By leveraging Conditional GANs to generate synthetic data that closely mirror real-world sensor readings, we effectively modeled various fault types, including hardover, drift, spike, erratic, and stuck faults. The transformation of time-series data into GAF images preserved temporal dependencies, while a CAE facilitated noise reduction and feature extraction. A lightweight ANN classifier subsequently achieved accurate fault categorization. Experimental results demonstrated a significant performance improvement, with the model achieving an accuracy of 95.93%, an $F1$-score of 95.84%, a precision of 95.87%, and a sensitivity of 95.88% when enriched with synthetic data. These findings underscore the transformative potential of integrating synthetic data generation and advanced feature extraction techniques for robust fault detection in WSNs.

Future work will explore decentralized and privacy-preserving learning paradigms, such as federated learning, to enhance data security and privacy by enabling on-device training at individual sensor nodes. In addition, the model will be adapted for real-time fault detection in dynamic and resource-constrained environments to ensure responsiveness and scalability. Advanced privacy-preserving techniques, including secure aggregation, will be integrated

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to further strengthen the system's scalability and resilience in large-scale distributed networks.

REFERENCES

[1] O. C. Abikoye et al., "Application of Internet of Thing and cyber physical system in Industry 4.0 smart manufacturing," in *Emergence of Cyber Physical System and IoT in Smart Automation and Robotics: Computer Engineering in Automation*. Cham, Switzerland: Springer, 2021, pp. 203–217.

[2] C. C. Sobin, "A survey on architecture, protocols and challenges in IoT," *Wireless Pers. Commun.*, vol. 112, no. 3, pp. 1383–1429, Jun. 2020.

[3] M. A. Ahmed et al., "LoRa based IoT platform for remote monitoring of large-scale agriculture farms in Chile," *Sensors*, vol. 22, no. 8, p. 2824, Apr. 2022.

[4] D. D. Dasig, "Implementing IoT and wireless sensor networks for precision agriculture," in *Internet of Things and Analytics for Agriculture, Volume 2*, (Studies in Big Data), vol. 67, P. Pattnaik, R. Kumar, and S. Pal, Eds., Singapore: Springer, 2020, doi: 10.1007/978-981-15-0663-5_2.

[5] P. Pawar, M. TarunKumar, and P. Vittal K., "An IoT based intelligent smart energy management system with accurate forecasting and load strategy for renewable generation," *Measurement*, vol. 152, Feb. 2020, Art. no. 107187.

[6] A. Afridi et al., "An efficient and improved model for power theft detection in Pakistan," *Bull. Electr. Eng. Informat.*, vol. 10, no. 4, pp. 1828–1837, Aug. 2021.

[7] U. Saeed, S. Y. Shah, J. Ahmad, M. A. Imran, Q. H. Abbasi, and S. A. Shah, "Machine learning empowered COVID-19 patient monitoring using non-contact sensing: An extensive review," *J. Pharmaceutical Anal.*, vol. 12, no. 2, pp. 193–204, Apr. 2022.

[8] S. Krishnamoorthy, A. Dua, and S. Gupta, "Role of emerging technologies in future IoT-driven healthcare 4.0 technologies: A survey, current challenges and future directions," *J. Ambient Intell. Humanized Comput.*, vol. 14, no. 1, pp. 361–407, Jan. 2023.

[9] U. Saeed et al., "Discrete human activity recognition and fall detection by combining FMCW RADAR data of heterogeneous environments for independent assistive living," *Electronics*, vol. 10, no. 18, p. 2237, Sep. 2021.

[10] A. Afridi, I. Hameed, C. E. García, and I. Koo, "Throughput maximization of wireless powered IoT network with hybrid NOMA-TDMA scheme: A genetic algorithm approach," *IEEE Access*, vol. 12, pp. 65241–65253, 2024.

[11] M. Ayaz, M. Ammad-Uddin, I. Baig, and E.-H.-M. Aggoune, "Wireless sensor's civil applications, prototypes, and future integration possibilities: A review," *IEEE Sensors J.*, vol. 18, no. 1, pp. 4–30, Jan. 2018.

[12] S. Javaid, H. Fahim, S. Zeadally, and B. He, "Self-powered sensors: Applications, challenges, and solutions," *IEEE Sensors J.*, vol. 23, no. 18, pp. 20483–20509, Sep. 2023.

[13] L. Xing, "Cascading failures in Internet of Things: Review and perspectives on reliability and resilience," *IEEE Internet Things J.*, vol. 8, no. 1, pp. 44–64, Jan. 2021.

[14] Y. Wei, D. Wu, and J. Terpenny, "Robust incipient fault detection of complex systems using data fusion," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 12, pp. 9526–9534, Dec. 2020.

[15] S. X. Ding, *Advanced Methods for Fault Diagnosis and Fault-Tolerant Control*. Cham, Switzerland: Springer, 2021.

[16] Y. Chi, Y. Dong, Z. J. Wang, F. R. Yu, and V. C. M. Leung, "Knowledge-based fault diagnosis in industrial Internet of Things: A survey," *IEEE Internet Things J.*, vol. 9, no. 15, pp. 12886–12900, Aug. 2022.

[17] J. Chen, L. Zhang, Y. Li, Y. Shi, X. Gao, and Y. Hu, "A review of computing-based automated fault detection and diagnosis of heating, ventilation and air conditioning systems," *Renew. Sustain. Energy Rev.*, vol. 161, Jun. 2022, Art. no. 112395.

[18] M. Nilashi, O. Ibrahim, H. Ahmadi, and L. Shahmoradi, "A knowledge-based system for breast cancer classification using fuzzy logic method," *Telematics Informat.*, vol. 34, no. 4, pp. 133–144, Jul. 2017.

[19] V. Sampath, I. Maurtua, J. J. A. Martín, and A. Gutierrez, "A survey on generative adversarial networks for imbalance problems in computer vision tasks," *J. Big Data*, vol. 8, no. 1, pp. 1–59, 2021.

[20] A. Dairi, F. Harrou, and Y. Sun, "Deep generative learning-based 1-SVM detectors for unsupervised COVID-19 infection detection using blood tests," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[21] S. Bouchenak, R. Merzougui, F. Harrou, A. Dairi, and Y. Sun, "A semi-supervised modulation identification in MIMO systems: A deep learning strategy," *IEEE Access*, vol. 10, pp. 76622–76635, 2022.

[22] N. Lakal, A. H. Shehri, K. W. Brashler, S. P. Wankhede, J. Morse, and X. Du, "Sensing technologies for condition monitoring of oil pump in harsh environment," *Sens. Actuators A, Phys.*, vol. 346, Oct. 2022, Art. no. 113864.

[23] A. Swain, E. Abdellatif, A. Mousa, and P. W. T. Pong, "Sensor technologies for transmission and distribution systems: A review of the latest developments," *Energies*, vol. 15, no. 19, p. 7339, Oct. 2022.

[24] R.-X. Guo, K. Guo, and J.-K. Dong, "Fault diagnosis for sensors in a class of nonlinear systems," *IMA J. Math. Control Inf.*, vol. 35, no. 2, pp. 375–391, 2018.

[25] W. Ahmad, S. A. Khan, M. M. M. Islam, and J.-M. Kim, "A reliable technique for remaining useful life estimation of rolling element bearings using dynamic regression models," *Rel. Eng. Syst. Saf.*, vol. 184, pp. 67–76, Apr. 2019.

[26] A. Mahapatro and P. M. Khilar, "Fault diagnosis in wireless sensor networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 4, pp. 2000–2026, 4th Quart., 2013.

[27] K. Islam, W. Shen, and X. Wang, "Wireless sensor network reliability and security in factory automation: A survey," *IEEE Trans. Syst., Man, Cybern., C, Appl. Rev.*, vol. 42, no. 6, pp. 1243–1256, Nov. 2012.

[28] S. U. Jan, Y.-D. Lee, J. Shin, and I. Koo, "Sensor fault classification based on support vector machine and statistical time-domain features," *IEEE Access*, vol. 5, pp. 8682–8690, 2017.

[29] X. Miao, Y. Liu, H. Zhao, and C. Li, "Distributed online one-class support vector machine for anomaly detection over networks," *IEEE Trans. Cybern.*, vol. 49, no. 4, pp. 1475–1488, Apr. 2019.

[30] M. Yoo, T. Kim, J. T. Yoon, Y. Kim, S. Kim, and B. D. Youn, "A resilience measure formulation that considers sensor faults," *Rel. Eng. Syst. Saf.*, vol. 199, Jul. 2020, Art. no. 106393.

[31] S. Zidi, T. Moulahi, and B. Alaya, "Fault detection in wireless sensor networks through SVM classifier," *IEEE Sensors J.*, vol. 18, no. 1, pp. 340–347, Jan. 2018.

[32] Z. Noshad et al., "Fault detection in wireless sensor networks through the random forest classifier," *Sensors*, vol. 19, no. 7, p. 1568, Apr. 2019.

[33] U. Saeed, S. U. Jan, Y.-D. Lee, and I. Koo, "Fault diagnosis based on extremely randomized trees in wireless sensor networks," *Rel. Eng. Syst. Saf.*, vol. 205, Jan. 2021, Art. no. 107284.

[34] U. Saeed, Y. D. Lee, S. U. Jan, and I. Koo, "CAFD: Context-aware fault diagnostic scheme towards sensor faults utilizing machine learning," *Sensors*, vol. 21, no. 2, p. 617, 2021.

[35] J. Huang, M. Li, Y. Zhang, L. Mu, Z. Ao, and H. Gong, "Fault detection and classification for sensor faults of UAV by deep learning and time-frequency analysis," in *Proc. 40th Chin. Control Conf. (CCC)*, Jul. 2021, pp. 4420–4424.

[36] Md. N. Hasan, S. U. Jan, and I. Koo, "Wasserstein GAN-based digital twin-inspired model for early drift fault detection in wireless sensor networks," *IEEE Sensors J.*, vol. 23, no. 12, pp. 13327–13339, Jun. 2023.

[37] R. Khan, S. U. Khan, U. Saeed, and I.-S. Koo, "Auscultation-based pulmonary disease detection through parallel transformation and deep learning," *Bioengineering*, vol. 11, no. 6, p. 586, Jun. 2024.

[38] N. Ijaz, F. Banoori, and I. Koo, "Reshaping bioacoustics event detection: Leveraging few-shot learning (FSL) with transductive inference and data augmentation," *Bioengineering*, vol. 11, no. 7, p. 685, Jul. 2024.

[39] U. Saeed et al., "Portable UWB RADAR sensing system for transforming subtle chest movement into actionable micro-Doppler signatures to extract respiratory rate exploiting ResNet algorithm," *IEEE Sensors J.*, vol. 21, no. 20, pp. 23518–23526, Oct. 2021.

[40] L. Gou, H. Li, H. Zheng, H. Li, and X. Pei, "Aeroengine control system sensor fault diagnosis based on CWT and CNN," *Math. Problems Eng.*, vol. 2020, pp. 1–12, Jan. 2020.

[41] T. Zhao, H. Zhang, X. Zhang, Y. Sun, L. Dou, and S. Wu, "Multi-fault identification of iron oxide gas sensor based on CNN-wavelelet-based network," in *Proc. 19th Int. Conf. Opt. Commun. Netw. (ICOCN)*, Aug. 2021, pp. 1–4.

[42] C. Li, J. Xiong, X. Zhu, Q. Zhang, and S. Wang, "Fault diagnosis method based on encoding time series and convolutional neural network," *IEEE Access*, vol. 8, pp. 165232–165246, 2020.

[43] H. Lu, M. Du, K. Qian, X. He, and K. Wang, "GAN-based data augmentation strategy for sensor anomaly detection in industrial robots," *IEEE Sensors J.*, vol. 22, no. 18, pp. 17464–17474, Sep. 2022.

[44] C. Lu, C. K. Reddy, P. Wang, D. Nie, and Y. Ning, "Multi-label clinical time-series generation via conditional GAN," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 4, pp. 1728–1740, Apr. 2024.

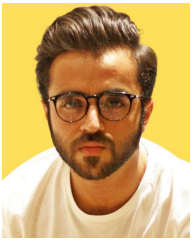
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[45] E. Zotov, A. Tiwari, and V. Kadirkamanathan, "Conditional StyleGAN modelling and analysis for a machining digital twin," *Integr. Comput.-Aided Eng.*, vol. 28, no. 4, pp. 399–415, 2021.
[46] S. U. Jan, Y. D. Lee, and I. S. Koo, "A distributed sensor-fault detection and diagnosis framework using machine learning," *Inf. Sci.*, vol. 547, pp. 777–796, Feb. 2021.
[47] R. Khan, U. Saeed, and I. Koo, "FedLSTM: A federated learning framework for sensor fault detection in wireless sensor networks," *Electronics*, vol. 13, no. 24, p. 4907, Dec. 2024.
[48] S. Suthaharan, M. Alzahrani, S. Rajasegarar, C. Leckie, and M. Palaniswami, "Labelled data collection for anomaly detection in wireless sensor networks," in *Proc. IEEE 6th Int. Conf. Intell. Sensors Sensor Netw. Inf. Process.*, Aug. 2010, pp. 269–274.
[49] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.
[50] Z. Wang and T. Oates, "Spatially encoding temporal correlations to classify temporal data using convolutional neural networks," 2015, *arXiv:1509.07481*.
[51] J. Ballé, V. Laparra, and E. P. Simoncelli, "End-to-end optimized image compression," 2016, *arXiv:1611.01704*.
[52] L. Theis, W. Shi, A. A. Cunningham, and F. Huszár, "Lossy image compression with compressive autoencoders," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2017, pp. 1–12.
[53] B. T. Atmaja, H. Ihsannur, and D. Arifianto, "Lab-scale vibration analysis dataset and baseline methods for machinery fault diagnosis with machine learning," *J. Vib. Eng. Technol.*, vol. 12, no. 2, pp. 1991–2001, Feb. 2024.



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